

Actively managed fund fee: 2% vs Index fund management fee: 0.05%

Passively managed fund account for 1/3 of the market share

"If a statue is ever erected to honor the person who has done the most for American investors, the handsdown choice should be John Bogle."
-- Buffet

John Bogle
The Founder of Index Fund
The Founder of Vanguard (with 5 trillion dollar under management)

If the market return is 7% and the manager generates exactly the market return, then 5% vs 6.95% for 50 years means $1 will turn into $10 vs $30

The purposes of asset pricing

- Select stocks
- evaluate a company's events
- Merge and acquicition
- evaluate the strategy of a company
- Pricing for a unlisted company（IPO）
- Equity incentive

How to make money

The Accounting Basis

Revenue	
Sales Revenue	$20,438
Operating Expenses	
Cost of goods sold	$7,943
Selling, general and administrative expenses	$8,172
Depreciation and amortization	$960
Other expenses	$138
Total operating expenses	$17,213
Operating income	**$3,225**
Non-operating income	$130
Earnings before Interest and Taxes (EBIT)	**$3,355**
Financial income	$45
Income before Interest Expense (IBIE)	**$3,400**
Financial expense	$190
Earnings before income taxes	**$3,210**
Income taxes	$1,027
Net Income	$2,183

项　　目
一、营业收入
减：营业成本
营业税金及附加
销售费用
管理费用
财务费用
资产减值损失
加：公允价值变动收益（损失以“-”号填列）
投资收益（损失以“-”号填列）
其中：对联营企业和合营企业的投资收益
二、营业利润（亏损以“-”号填列）
加：营业外收入
减：营业外支出
其中：非流动资产处置损失
三、利润总额（亏损总额以“-”号填列）
减：所得税费用
四、净利润（净亏损以“-”号填列）
五、每股收益：

How to allocate

Adjust the income statement

EXHIBIT 1.29 Adjusted Income Statement

($ in millions, except per share data)

Income Statement	Reported 2007	Adjustments +	Adjustments −	Adjusted 2007
Sales	**$1,000.0**			**$1,000.0**
Cost of Goods Sold	625.0		(5.0)	620.0
Gross Profit	**$375.0**			**$380.0**
Selling, General & Administrative	230.0			230.0
Restructuring Charges	10.0		(10.0)	-
Operating Income (EBIT)	**$135.0**			**$150.0**
Interest Expense	35.0			35.0
Pre-tax Income	**$100.0**			**$115.0**
Income Taxes	40.0	6.0		46.0
Net Income	**$60.0**			**$69.0**
Operating Income (EBIT)	$135.0	15.0		$150.0
Depreciation & Amortization	50.0			50.0
EBITDA	**$185.0**			**$200.0**
Weighted Avg. Diluted Shares	30.0			30.0
Diluted EPS	$2.00			$2.30

Inventory write-down

Restructuring charge related to severance from downsizing the sales force

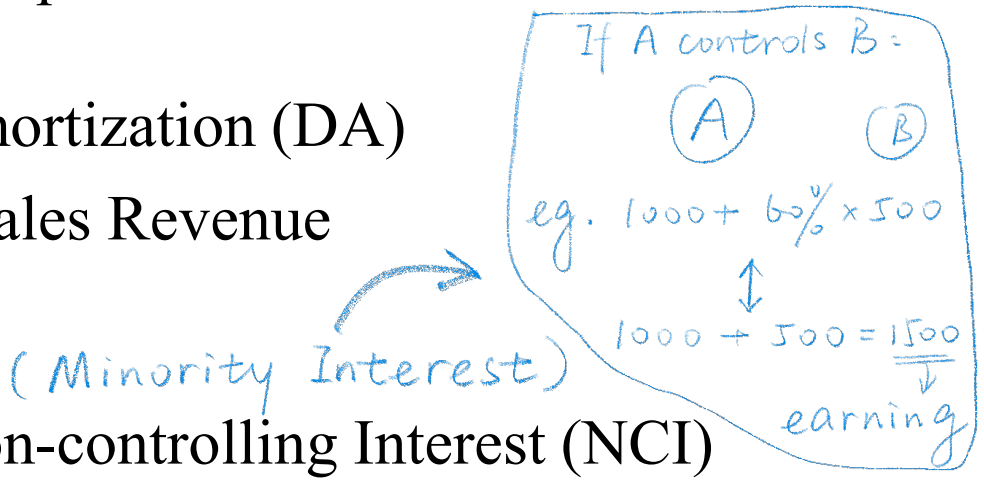
= (Inventory write-down + Restructuring charge) x Marginal Tax Rate
= ($5.0 million + $10.0 million) x 40.0%

D&A is typically sourced from a company's cash flow statement although it is sometimes broken out on the income statement

$15.0 million add-back of total non-recurring items

The Accounting Basis

- **Some important financial data**
  - Earnings before Interest, Taxes, Depreciation and Amortization (EBITDA)
    - » =EBIT + Depreciation and amortization (DA)
    - » EBITDA Margin=EBITDA/Sales Revenue
  - Net Income
    - » =EBIT – Interest– Taxes – Non-controlling Interest (NCI)
      - – Preferred Dividend
    - » It is customary to calculate net income as earning per share (EPS)
    - » P/E ratio = Share price/ EPS
    - » ROE=Net Income/Net Asset
    - » ROA=Net Income/Total Asset



The Accounting Basis

- **Some important financial data**
  - –enterprise value (EV)
    - » =stock market value+ total liability +preferred dividend + non-controlling interest –value of associate company - cash and cash equivalents

  A simplified version: EV= Stock market value+ Total liability -cash and cash equivalents

  - –Interest coverage ratio
    - » =EBIT、EBITDA /Interest

Asset Price

- Present Value of all Capitals：

$$PV = \frac{CF_1}{(1+k)} + \frac{CF_2}{(1+k)^2} + \frac{CF_n}{(1+k)^n}$$

$$= \sum_{t=1}^{n} \frac{CF_t}{(1+k)^t}$$

- If CF remains unchanged：$PV = \frac{CF_1}{k}$

- If CF is growing at a rate g ：

$$PV = \frac{CF_1}{k-g}$$

Asset Price

- CF Unchanged $$PV = \frac{CF}{k}$$
- CF grows at a rate g $$PV = \frac{CF}{k - g}$$
- Suppose CF of a company equals 100 each year，discounted rate =20%，then:

  If g=0, PV=100/0.20=500

  g=5%， PV=100/0.15=667

  g=18%，PV=100/0.02=5000

  g=19%，PV=100/0.01=10000

  g=19.5%，PV=100/0.005=20000

Estimate Value are Extremely Sensitive to Growth Rate.

CAPM

- Generally Speaking, for any security i,

$$E(r_i) - r_f = \beta_i \left[E(r_M) - r_f \right]$$

$$\text{where } \beta_i = \frac{Cov(r_i, r_M)}{\sigma_M^2}.$$

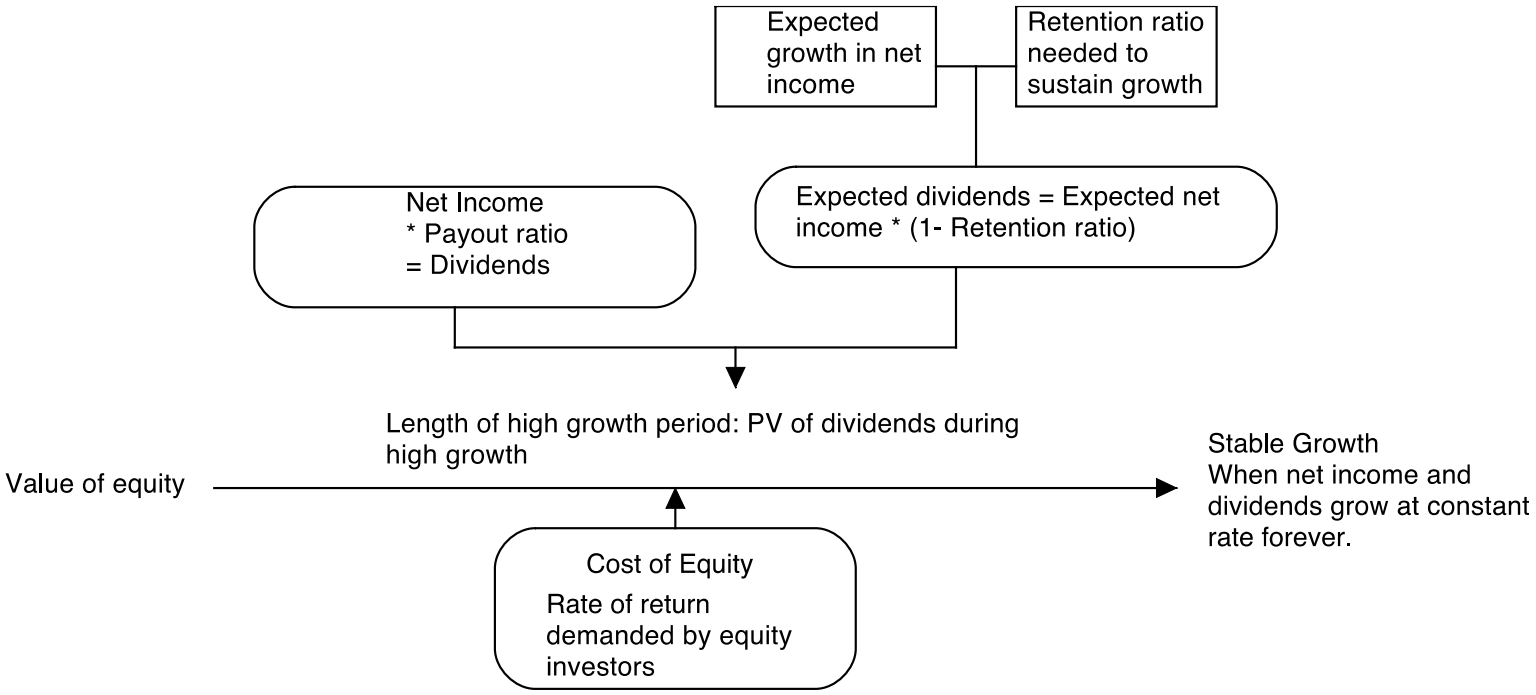
- This model demonstrates the relationship of expected rate of return between particular securities and whole market;

Example: Assume the risk free rate is 4%, the stock market average return is 11%. Firm A's Beta is 1.2.

According to the CAPM theory, the expected return (or the discount factor, or cost of equity) of firm A should be

R=4%+1.2*(11%-4%)=12.4%

Start easy: The Dividend Discount Model



The dividend discount model is the oldest DCF model around and it reflects the fact that bonds preceded equities into financial markets. Thus, investors in publicly traded stocks initially considered them to be "special" bonds, with the dividends replacing coupons and an uncertain (and appreciating) principal replacing the face value of the bond.

Dividend discounted Model

- Gordon Model, also known as Constant Growth Model, assumes that dividend on shares are increasing on a constant rate of g:
  - （r >g）

$$V_0 = \frac{D_0(1+g)}{(1+r)} + \frac{D_0(1+g)^2}{(1+r)^2} + \ldots = \frac{D_0(1+g)}{r-g} = \frac{D_1}{r-g}$$

The biggest weakness of the dividend discount model is not that it cannot value stocks that don't pay dividends but that it misvalues companies that pay out more or less than they can afford to in dividends.

Example：Constant Growth Model (CGM)

- This is 2009. In 2008, American Electric Power (AEP) paid a dividend of $1.64 per share. From 2006 to 2008, the average dividend increased 3.3% a year.
- From 1927 to 2008, the average market return in U.S was 11.4%. The historical average risk-free rate is 6.4%.
  According to the data, a valuation of the market risk premium is 5%.
- AEP's beta is 0.62 (defensive stock). The yield of 10-year bond is 3.32%..
- What's the discounted rate?
  – r=3.32%+0.62*5%=6.42%.
- What is the valuation?

$$V = \frac{1.64 \times 1.033}{0.0642 - 0.033} = 54.30$$

- The price was $31.06 on September 25, 2009.

Applicable conditions & limitations of CGM

- Applicable conditions
  - Mature companies that pay stable dividends.
    - » Dividends can represent cash flow.
  - A company whose growth rate is close to or below economic growth.
  - A company that is not cyclical.

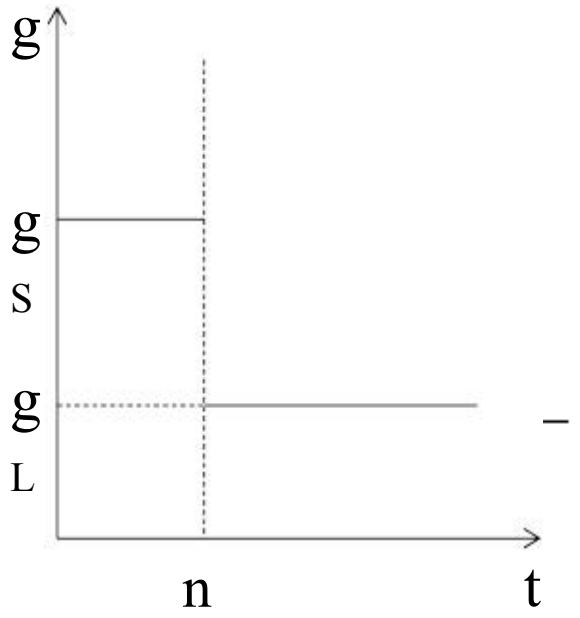
- Limitations
  - Very sensitive to growth rates.
  - Undervalue companies which scarcely pay dividend.
  - Not applicable to unstable companies, cyclical industries.

Two-stage Dividend Growth Model

Two-stage model assumes that the company's dividend growth experienced two stages.

In the first stage, dividend increases every year by a higher (or lower) growth rate.

After that time, growth of dividend becomes stable at a low (or high) level (as shown on the right) :



Two-stage Dividend Growth Model

Two-stage Dividend Growth Model

$$P_0 = \sum_{1}^{n} \frac{D_t}{(1+k_{hg})} + \frac{P_n}{(1+k_{hg})^n}$$

$$P_0 = \frac{D_0 \times (1+g) \times \left[1 - \frac{(1+g)^n}{(1+k_{hg})^n}\right]}{(k_{hg} - g)} + \frac{D_0 \times (1+g)^n \times (1+g_n)}{(k_{st} - g_n) \times (1+k_{hg})^n}$$

g = the growth rate of the first n years

k_{hg} = discounted rate of high growth period

k_{st} = discounted rate of stable growth period

g_n = stable growth rate after n years

The Choice of Long-term Growth Rate

- Long-term reinvestment ratio b, dividend growth rate g and ROE must be consistent.

$$g = b \times ROE$$

(26')

$ROE = \frac{NI}{BE}$

$Dividend = NI \times (1-b)$

$BE_{n+1} = BE_n + NI \cdot b$

book value of the debt / book value of the equity $= \frac{BD}{BE} = L$ is constant

$g = \frac{NI \cdot b}{BE} = ROE \cdot b$

$A_n = BE_n + BD_n$

$A_{n+1} = A_n \cdot \left(1 + \frac{NI \cdot b}{BE}\right)$

$NI_{n+1} = \left(1 + \frac{NI_n \cdot b}{BE_n}\right) \cdot NI_n$

Hidden Assumptions:

(1) The revenue/earnings is proportional to the total asset;
(2) The firm leverage is constant

Case：Consolidated Edison

One of the largest energy companies in the U.S.
About $14 billion in sales.
$36 billion in assets.

book value



Pricing is required in August 2008.
Last 12 months：
EPS：$3.17
DPS：$2.32
Beta：0.8
The U.S. 10-year Treasury rate：4.1%
US market risk premium：4.5%
Companies' leverage：70% equity，30% debt, stable for several decades.

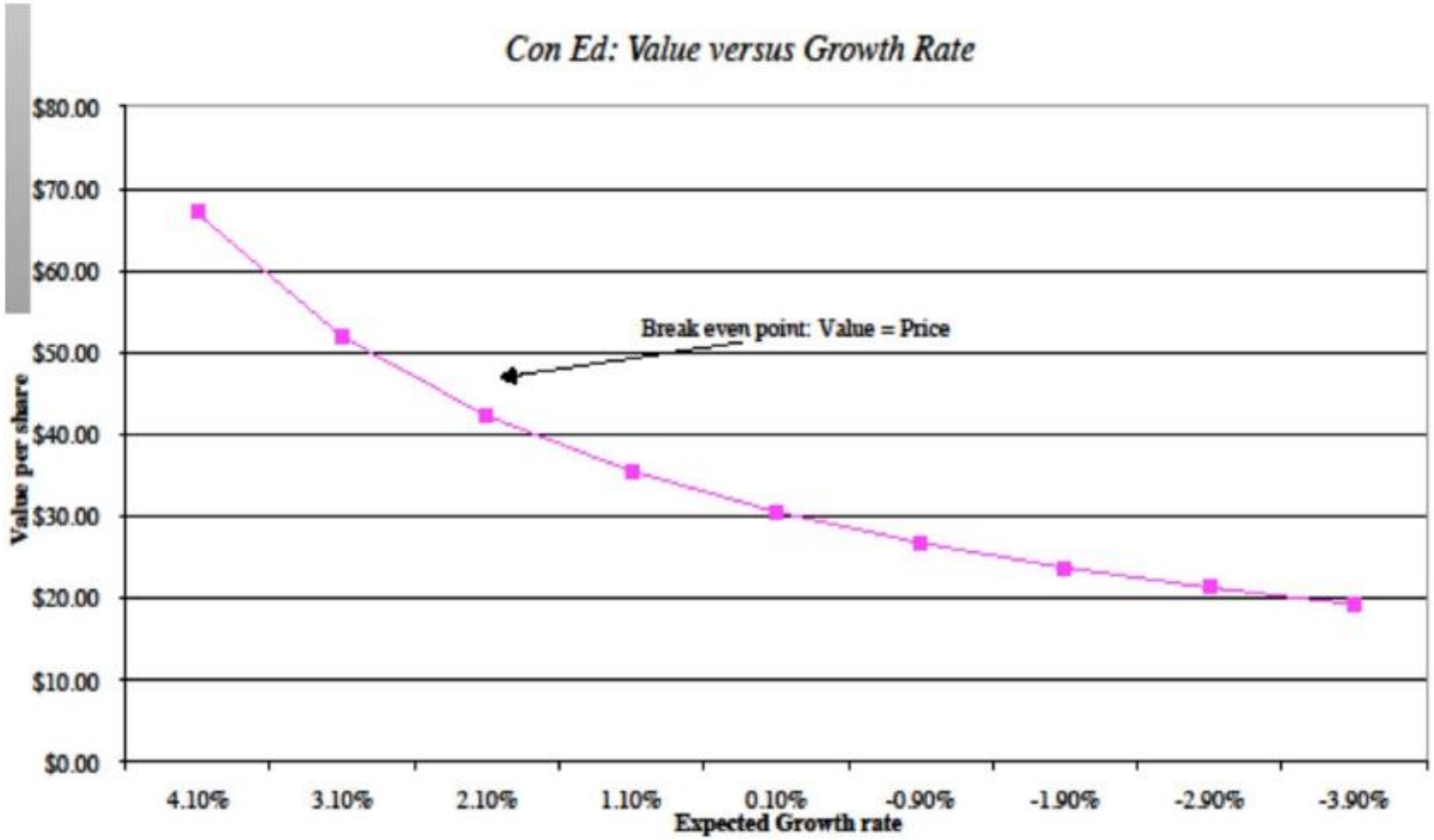
Which Model should We Choose?

- Why choose the simple dividend discount model?
  - Stable growth
    - » Regulated utility companies, with limited growth, are subject to population.
  - Pricing for what company?
    - » Leverage is very stable.
  - Dividend
    - » The company has paid out 97% of its free cash flow in dividends over the past five years.

Pricing

- Discount rate
  - =4.1%+0.8*4.5%=7.7%
- Stable discount rate
  - b=1-DPS/EPS=27%
  - ROE=7.7%
  - g=ROE*b=2.1%
- Price
  - =2.32*(1.021)/(0.077-0.021)
  - =42.30
- August 12, 2008, the market price is $40.76.

Price &Growth Rate



Case：ABN AMRO

The largest bank in the Netherlands
Total asset is about 600 billion euros

In December of 2003
EPS：1.85 euro
DPS：0.90 euro
European Banks' average beta at the time：0.95
Euro long-term risk-free interest rate：4.35%
Mature market's risk premium：4%
ROE: 16%



Pricing

- Consider applying two-stage model due to extremely high ROE:
- The Discount Rate
  - =4.35%+0.95*4%=8.15%
- The First Stage
  - b=(1-0.9/1.85)=51.35%
  - ROE=16%
  - g=51.35%*16%=8.22%
- The Second Stage
  - g=4%
  - Discount Rate=4.35%+1*4%=8.35%
  - ROE=8.35%
  - Dividend Rate=1-4/8.35=0.521

Pricing

- Terminal Value

$$EPS_6 = 1.85 \times (1.0822)^5 \times 1.04 = 2.86$$

$$P_5 = EPS_6 \times 0.521 / (0.0835 - 0.04) = 34.20$$

- Price

$$\frac{EPS \times (1-b_{hg}) \times (1+g_{hg}) \times \left(1 - \frac{(1+g_{hg})^5}{(1+k_{hg})^5}\right)}{k_{hg} - g_{hg}} + \frac{P_5}{(1+k_{hg})^5}$$

$$= \frac{1.85 \times 48.65\% \times (1+8.22\%) \times \left(1 - \frac{(1+8.22\%)^5}{(1+8.15\%)^5}\right)}{8.15\% - 8.22\%} + \frac{34.20}{(1+8.15\%)^5}$$

$$= 27.62$$

- In December 2003： market price was 18.55 Euro
  End of the year 2007： RBS offered 38.5 Euro per share to takeover ABN AMRO

Model Weakness

Defect

– How long will high growth rate last?

– Growth rate will fall suddenly.

A Method to Adjust Free Cash Flow

- American companies are more likely paying dividends by repurchasing shares.
  - Now, over 50% of dividends are paid by repurchasing shares.
    - » Why?
- An adjusting method:
  - "Dividend" =dividend + Buy-Back
  - Dividend Ratio=（Dividend + Share Repurchase) / net earnings

(略…至P33)

Case：S&P500 Index（S&P500:January, 2009）

The dividend during last 12 months：28.05
Estimated value of g = 4% in next 5 years

Treasury Bond Long-term Risk-free Rate：2.21%
Current Market Risk Premium：6%

Pricing

- Consider applying two stage model
  - Current expected growth rate is much greater than current interest rate
- Discount rate
  - =2.21%+1*6%=8.21%
- First Stage
  - g=4%
- Last Stage
  - g=2.21%
  - Assuming market risk premium gets back to 4%

Pricing

- Result
  - Present value of high growth stage:124.71
  - Terminal Value：587.75
  - Value：712.46
- On January 1st 2009：
  - Actual Index Price：903
- Any questions?

Pricing

- Adjust Dividend pricing Model, plus repurchasing shares
- Dividends during last 12 months + repurchasing Shares = 52.58

- Result
  - Present value of high growth stage: 233.76
  - Terminal Value： 1101.75
  - Value: 1335.51

Case：S&P500 Index （S&P500：January, 2010）

Dividends during last 12 months + Repurchasing Shares ： 40.38
Estimated value of g = 7.2% in next 5 years

Treasury Bond Long-term Risk-free Rate ： 3.84%
Current Market Risk Premium ： 5%

Pricing

- Considering two stage model
  - At this stage the expected growth rate is far higher than interest rates.
- discounted rate
  - =3.84%+1*5%=8.84%
- The first stage
  - g=7.2%
- The last stage
  - g=3.84%
  - Assume the market risk premium goes back to 4%

Pricing

- The result
  - The present value of high growth stage： 192.95
  - The present value of ultimate stage ： 971.63
  - The price： 1164.59

- 2010.1.1
  - The actual point of the index： 1115

Three-Stage model

- The high growth stage were divided into high growth and middle growth stage, the last is the stable growth phase
  - the transformation of stable growth phase and the middle growth stage can be little by little.

Case：Goldman Sachs：2008.8

2008.8
EPS：16.77 dollar
DPS：1.40 dollar
Average Beta of Investment Bank: 1.40
Risk free rate of T-Bills：4.10%
Risk premium of mature market：4.5%
ROE: 13.19%



Pricing

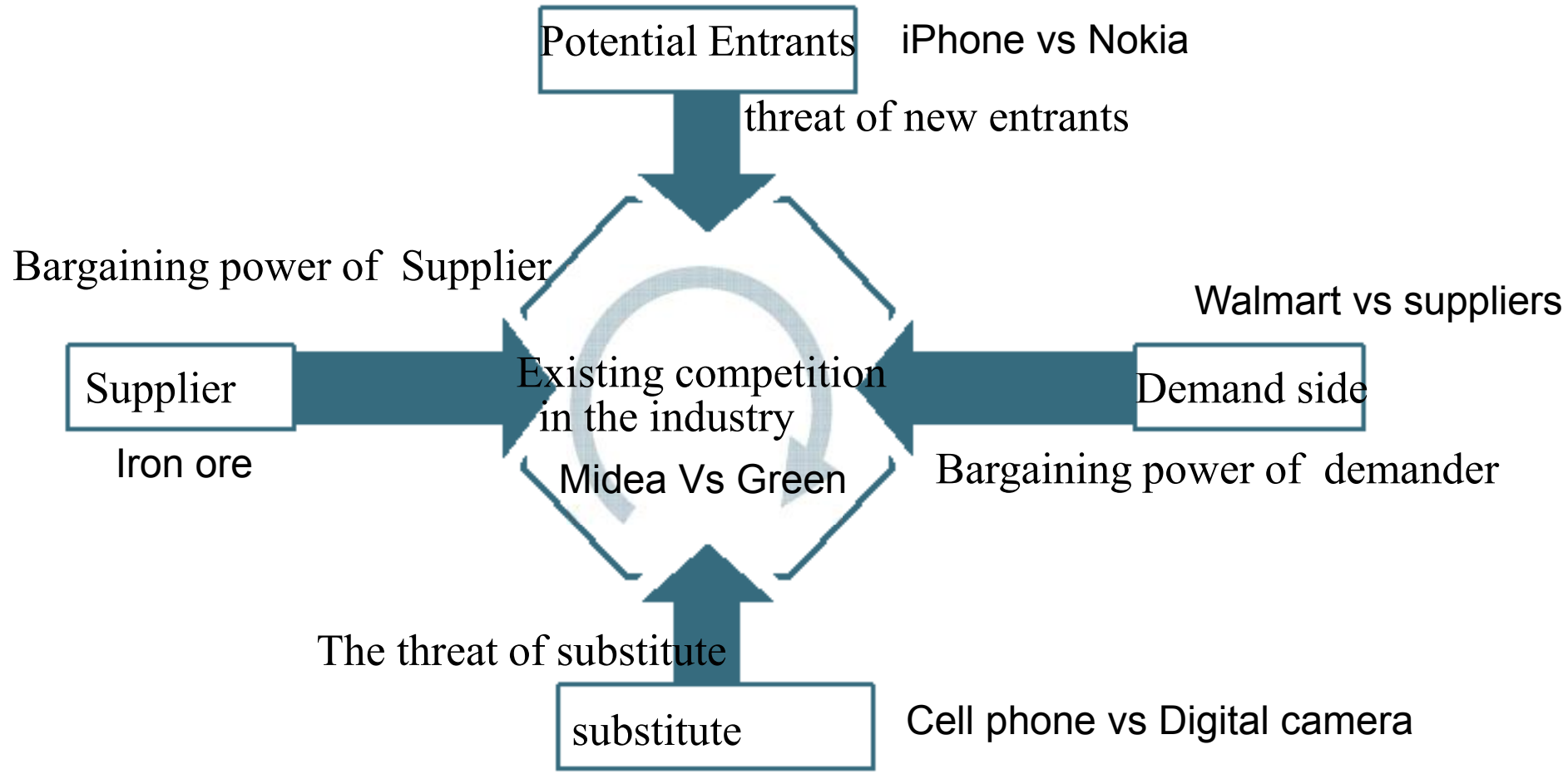
- Considering three-stage model
  - The dividend payout and the growth pattern of the company shows the potential of the company.
- discounted Rate
  - =4.10%+1.40*4.5%=10.4%
- The first stage
  - b=（1-1.4/16.77)=91.65%
  - ROE=13.19%
  - g=91.65%*13.19%=12.09%
- The last stage
  - g=4%
  - discounted rate=4.10%+1.2*4.5%=9.5%
  - ROE=10%

Pricing

- The secondary stage
  - growth rate、dividend payout ratio and beta are all gradually move to stable stage in the period from year 6 to 10.
  - Price after calculation=222.49 dollar
  - 2008.8:the price of Goldman Sachs 169 dollar

What factors decide the earning growth rate?

The Five-force Model



Michael Porter

- In January 2013, at Harvard business school, professor Porter awarded the Porter Award to outstanding Japanese firms. In less than two months, his own business consulting firm Monitor went into bankruptcy. Professor Porter's five factor model has been widely taught in business schools for generations of MBA students. However, it seems that the model is more appropriate to explain history, rather than to predict future. A firm’s long term performance depends on its innovation capacity, rather than shielding competition.



What is a good company ?

- How profitable it should be for a good company?

- How many years can the high profitability last?

What is a good company： American data

	percentage								
	2%	10%	25%	40%	50%	60%	75%	90%	98%
One-year sales growth	-36.4	-2.4	5.7	9.3	11.3	13.3	17	25.2	47.7
ROE	-67.5	-25.3	-2.8	6.9	11	14.9	23.1	45.9	216.6

What's a good company: American data

Firms persistently outperform the median level

Year	1	2	3	4	5	6	7	8	9	10
Sales growth rate										
percentage	50	30	18.6	11.9	7.8	5.6	3.4	2.4	1.5	1.2
ROE										
percentage	50	25.6	13	6.8	3.6	2	1.2	0.7	0.5	0.3

What is a good company: American data

	Firms persistently beat the median									
	1	2	3	4	5	6	7	8	9	10
					Big firm					
Sales growth	53	31.3	18.9	11.7	7.5	4.8	3.2	2.2	1.6	1.1
ROE	46.7	21.9	10	4.7	2.2	1.2	0.7	0.4	0.3	0.2
					Small firm					
Sales growth	47	26.1	14.7	8.6	5.2	3.2	2.1	1.4	1	0.7
ROE	53.3	25	12.5	6.3	3.1	1.6	0.8	0.4	0.2	0.1

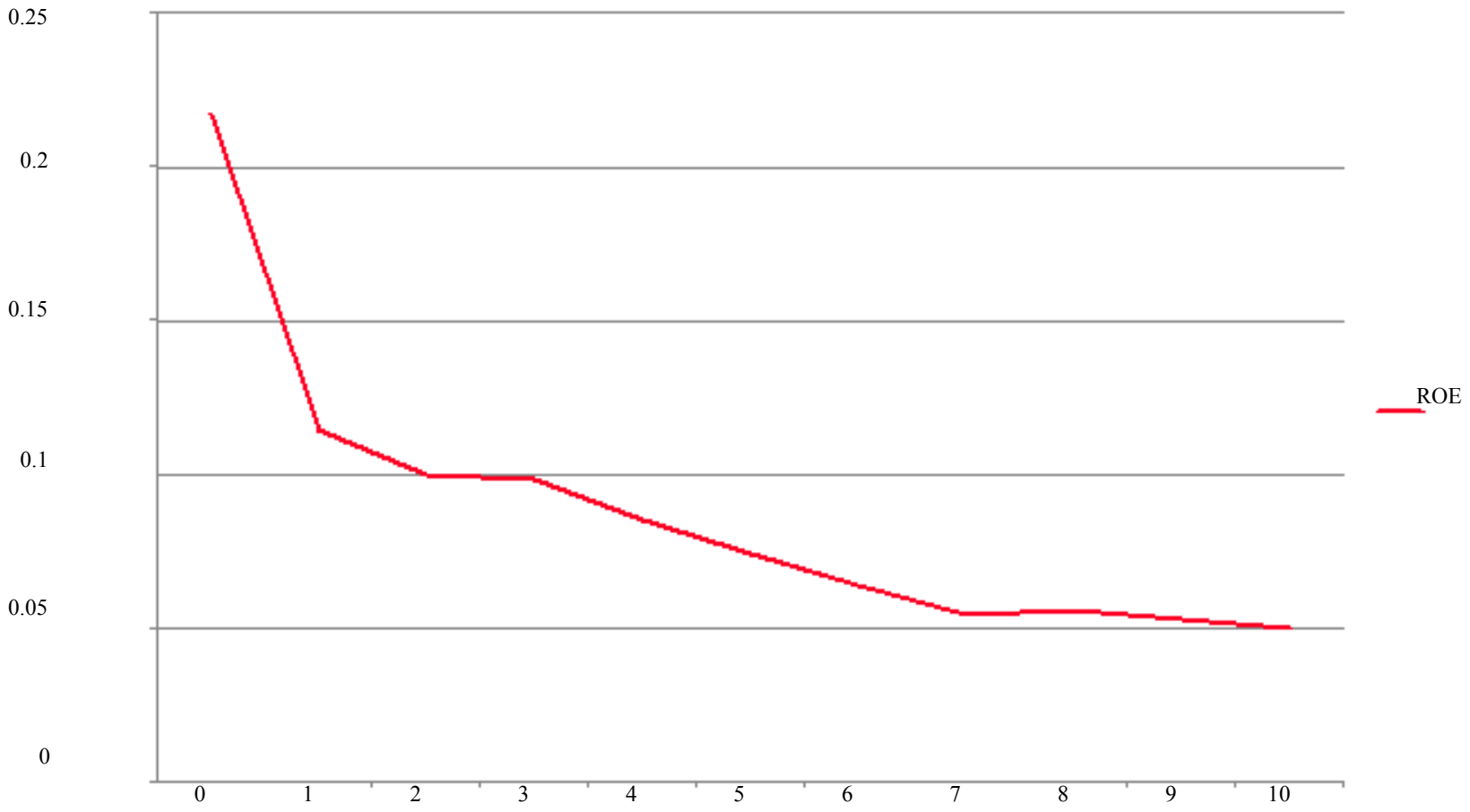
- Do not show growth potential
- Expensive

	ROE	PB	PE
Before IPO	0.25		76
After IPO	0.10	3.88	57

Money usage in 创业板

- 2011年5月3日深交所报告
  - 创业板209家上市公司IPO融资金额1618亿元，超募资金1002亿元
  - 创业板公司平均每家IPO融资达7.69亿元，其中超募4.79亿元，一年半下来，真正用于与主业相关的投资项目比例还不到20%。

ROE and listed years in China



创业板中报大变脸

近一年创业板上市业绩预减公司一览

证券代码	证券简称	上市公告日	预告同比增幅下限（%）	预告同比增幅上限（%）	首发主承销商	摊薄首发市盈率（倍）	首发超募资金（万元）
300241.SZ	瑞丰光电	2011年7月11日	-12.00	0.00	华龙证券	26.54	2465
300242.SZ	明家科技	2011年7月11日	-50.00	-40.00	民生证券	37.04	4324
300243.SZ	瑞丰高材	2011年7月11日	-20.00	0.00	平安证券	22.54	1386
300245.SZ	天玑科技	2011年7月18日	-29.79	——	安信证券	34.13	13002
300246.SZ	宝莱特	2011年7月18日	-30.00	0.00	平安证券	31.25	10900
300247.SZ	桑乐金	2011年7月28日	-35.00	-10.00	平安证券	29.63	12629
300250.SZ	初灵信息	2011年8月2日	-30.00	0.00	国信证券	32.89	11563
300252.SZ	金信诺	2011年8月17日	-20.00	0.00	中航证券	32.73	11999
300254.SZ	仟源制药	2011年8月18日	-49.31	-40.87	中原证券	43.33	18331
300256.SZ	星星科技	2011年8月18日	-70.00	-50.00	国信证券	39.62	-10616
300260.SZ	新莱应材	2011年9月5日	-29.00	-24.00	长江证券	46.63	9345
300262.SZ	巴安水务	2011年9月15日	-40.00	-30.00	平安证券	39.13	8964
300263.SZ	隆华传热	2011年9月15日	-20.00	0.00	光大证券	55.00	36676
300268.SZ	万福生科	2011年9月26日	-17.25	-14.36	平安证券	33.78	1556
300269.SZ	联建光电	2011年10月11日	-25.00	0.00	东方证券	38.31	14101
300274.SZ	阳光电源	2011年11月1日	-30.00	-15.00	国元证券	41.48	94489
300276.SZ	三丰智能	2011年11月14日	-15.00	0.00	海通证券	45.54	14270
300279.SZ	和晶科技	2011年12月28日	-20.00	0.00	申银万国	31.33	10614
300280.SZ	南通锻压	2011年12月28日	-25.00	-15.00	海通证券	34.38	647
300283.SZ	温州宏丰	2012年1月9日	-80.00	-60.00	中德证券	32.32	1534
300292.SZ	吴通通讯	2012年2月28日	-45.00	-35.00	东吴证券	23.67	-836
300307.SZ	慈星股份	2012年3月28日	-40.00	-30.00	国信证券	16.43	12392
300308.SZ	中际装备	2012年4月9日	-5.00	-0.10	广发证券	26.73	9210
300313.SZ	天山生物	2012年4月24日	-15.00	-5.00	民生证券	56.52	14657
300316.SZ	晶盛机电	2012年5月10日	-20.00	0.00	国信证券	34.74	55886
300317.SZ	珈伟股份	2012年5月10日	-95.85	-94.12	国泰君安	27.50	-7737
300319.SZ	麦捷科技	2012年5月22日	-25.00	-10.00	平安证券	27.82	5727
300323.SZ	华灿光电	2012年5月31日	-42.80	-34.60	中信证券	34.42	2256
300324.SZ	旋极信息	2012年6月7日	-20.00	-5.00	平安证券	36.99	21115
300327.SZ	中颖电子	2012年6月12日	-38.30	-28.90	国信证券	26.60	14560
300330.SZ	华虹计通	2012年6月18日	-24.59	-22.67	东方证券	35.71	16064

资料来源：Wind资讯

Royal Dutch Shell Feb 21, 2020

Royal Dutch Shell plc (RDS-A)
NYSE - NYSE Delayed Price. Currency in USD

Add to watchlist

49.17 -0.45 (-0.91%)
At close: 4:02PM EST

Buy Sell

Summary | Company Outlook | Chart | Conversations | Statistics | Historical Data | Profile | Financials | Analysis | Options

Previous Close	49.62	Market Cap	183.648B
Open	49.06	Beta (5Y Monthly)	0.82
Bid	49.07 x 2200	PE Ratio (TTM)	12.61
Ask	49.32 x 2200	EPS (TTM)	3.90
Day's Range	48.76 - 49.31	Earnings Date	N/A
52 Week Range	48.76 - 66.48	Forward Dividend & Yield	3.76 (7.58%)
Volume	5,471,043	Ex-Dividend Date	Feb 12, 2020
Avg. Volume	3,710,129	1y Target Est	73.29



Shell had $20 billion cash reserve

Google Feb 21, 2020

Alphabet Inc. (GOOG)
NasdaqGS - NasdaqGS Real Time Price. Currency in USD

Add to watchlist

1,485.11 -33.04 (-2.18%)
At close: 4:00PM EST

Buy Sell

Summary | Company Outlook | Chart | Conversations | Statistics | Historical Data | Profile | Financials | Analysis | Options

Previous Close	1,518.15	Market Cap	1.02T
Open	1,508.03	Beta (5Y Monthly)	1.04
Bid	1,481.73 x 1000	PE Ratio (TTM)	30.21
Ask	1,485.99 x 1200	EPS (TTM)	49.16
Day's Range	1,480.44 - 1,512.21	Earnings Date	N/A
52 Week Range	1,025.00 - 1,532.11	Forward Dividend & Yield	N/A (N/A)
Volume	1,632,854	Ex-Dividend Date	N/A
Avg. Volume	1,457,486	1y Target Est	1,616.37



Google had $117 billion cash in hand!

Apple Feb 21, 2020

Apple Inc. (AAPL)
NasdaqGS - NasdaqGS Real Time Price. Currency in USD

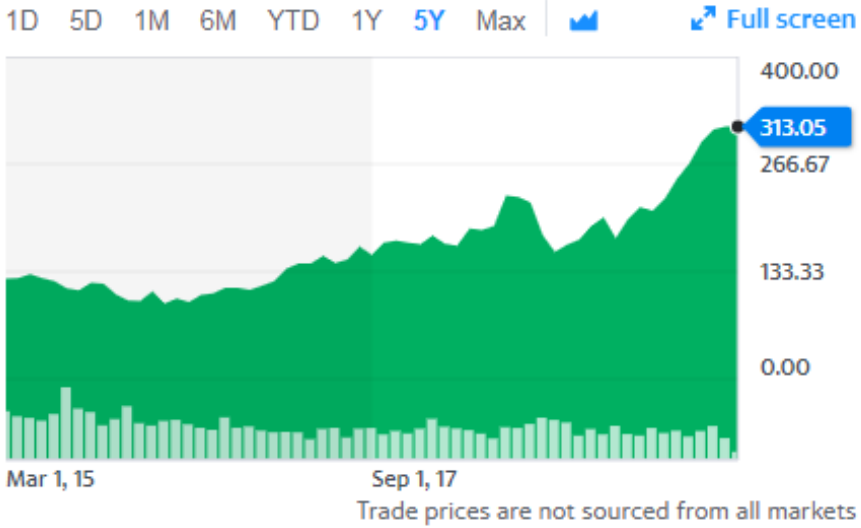
Add to watchlist

313.05 -7.25 (-2.26%)
At close: 4:00PM EST

Buy | Sell

Summary | Company Outlook | Chart | Conversations | Statistics | Historical Data | Profile | Financials | Analysis | Options

Previous Close	320.30	Market Cap	1.37T
Open	318.62	Beta (5Y Monthly)	1.28
Bid	312.10 x 3000	PE Ratio (TTM)	24.86
Ask	312.38 x 900	EPS (TTM)	12.60
Day's Range	310.50 - 320.42	Earnings Date	Apr 27, 2020 - May 03, 2020
52 Week Range	169.50 - 327.85	Forward Dividend & Yield	3.08 (0.96%)
Volume	32,426,415	Ex-Dividend Date	Feb 06, 2020
Avg. Volume	29,963,150	1y Target Est	333.31



Apple had $207 billion cash in hand!

The GDP of South Africa is 337.6 billion USD in 2020, with population 58 million
The GDP of Malaysia is 337.3 billion USD in 2020, with population 33.2 million

Market Capitalization Management and Valuation

- Why investing?

$$\frac{P}{E} = \frac{(1-b)}{r - b \times ROE}$$

$$\frac{P}{B} = \frac{ROE\ (1-b)}{r - b \times ROE}$$

$\Leftarrow$

$P_t = \frac{D_{t+1}}{r-g}$

$g = ROE \cdot b \qquad D_{t+1} = E_{t+1} \cdot (1-b)$

$P_t = \frac{E_{t+1}(1-b)}{r - ROE \cdot b}$

$\frac{P_t}{E_{t+1}} = \frac{1-b}{r - ROE \cdot b}$

$\frac{P_t}{B_{t+1}} = \frac{P_t}{E_{t+1}} \cdot \frac{E_{t+1}}{B_{t+1}} = \frac{1-b}{r - ROE \cdot b} \cdot ROE$

- How much should investment rate b be to create value?

$$\frac{d(P/E)}{db} = \frac{-1 \cdot (r - ROE \cdot b) + ROE(1-b)}{(r - ROE \cdot b)^2} = \frac{ROE - r}{(r - ROE \cdot b)^2}$$

if $ROE > r$: $b = 1$

$ROE = r$: $P_t/E_{t+1} \Rightarrow P/E = \frac{1}{r}$ whatever b

$ROE < r$: $b = 0$

公司决策：dividend/repurchase

Market Capitalization Management and Valuation

- When to invest?
  - the investment（b）would decrease the cash flow（1-b）
    - » with all other factors the same, firms with heavy asset are inferior to firms with light asset
  - The investment only creates value when the ROE is high enough

- Critical point：
  - ROE=r： cash cow
  - ROE>r: high growth company
  - ROE<r： investment ruins the wealth
    - » the market value drops below book value

$$\frac{P}{E} = \frac{(1-b)}{r - b \times ROE}$$

Market Capitalization Management and Valuation

- The high growth on company earnings is hard to maintain, even if it can, it is not necessary create true value
  - ROE may be too low (lower than cost of capital)
    - » the weakness of business model
  - Many companies have low profit margin, but they keep issuing new shares to finance new capital expenditure.
  - IPO provides a platform for many companies to ruin the value of their shareholders

Market Capitalization Management and Valuation

- Example of wealth being ruined
  - Photovoltaic enterprise：尚德、赛维
  - Metallurgy industry in China

- Should a company always expand?
  - depends on whether it creates wealth
  - If not
    - » split, strip
      GE

Why companies do not pay dividend?

Why companies do not pay dividend?

- the need for future investment
- risk control
- tax rate
- the signal for long-term profitability
- management benefit

Return on Input Capital (ROIC) measures the efficiency of capital utility

ROIC=Net Operating Profit after Tax (NOPAT) / (Average Total Invested Capital)
Average Total Invested Capital=Debt+Equity-Cash-Intangible asset

NOPAT				
($ in millions)			Company A	Company B
Revenue			$300	$300
Less: COGS			(50)	(50)
Gross Profit			**$250**	**$250**
Less: SG&A			(40)	(40)
Operating Income (EBIT)			**$210**	**$210**
Less: Interest Expense			--	(100)
Pre-Tax Income (EBT)			**$210**	**$110**
Less: Taxes	% Rate	35.0%	(74)	(39)
Net Income			**$137**	**$72**

NOPAT Approach #1		
Operating Income (EBIT)	$210	$210
Less: Taxes	(74)	(74)
NOPAT	**$137**	**$137**

NOPAT Approach #2		
Net Income	$137	$72
Plus: Interest Expense	--	100
Plus: Taxes	74	39
Operating Income (EBIT)	**$210**	**$210**
Less: Taxes	(74)	(74)
NOPAT	**$137**	**$137**

If ROIC<WACC, the firm is wasting capital.

The Value of Growth

- If dividends are omitted, the growth value comes from

$$\frac{P_2}{P_1} = \frac{P_2/E_2}{P_1/E_1} \times \frac{E_2}{E_1}$$

$$= \frac{P_2/E_2}{P_1/E_1} \times \frac{ROE_2}{ROE_1} \times \frac{B_1}{B_0}$$

P : market value

E: earnings

B: book value

$ROE_1 = E_1/B_0$

The Value of Growth

- Example 1：
  - first round of start-up business：
    B=100, E=20，ROE=20%，PE=6, Market value=20*6=120；
  - assume the similar company has PE equal to 10
    - » a year later, the company's market value goes to 240

$$\frac{P_2}{P_1} = \frac{P_2 / E_2}{P_1 / E_1} \times \frac{ROE_2}{ROE_1} \times \frac{B_1}{B_0} = \frac{10}{6} \times \frac{20\%}{20\%} \times \frac{120}{100} = 2$$

» conclusion： value created by valuation and capital accumulation

The Value of Growth

- Example 2：
  - first round of start-up business:
    B=100, E=20, ROE=20%, PE=120/20=6；
  - assume that by leverage on the first year， ROE=40%
  - assume the similar company has PE equal to 8 (lower than without leverage)
    - » a year later, create a company with market value of 384

$$\frac{P_2}{P_1}=\frac{P_2/E_2}{P_1/E_1}\times\frac{ROE_2}{ROE_1}\times\frac{B_1}{B_0}=\frac{8}{6}\times\frac{40\%}{20\%}\times\frac{120}{100}=3.2$$

  - » conclusion： value created by valuation, earnings and leverage

The Value of Growth

- Example 3：
  - first round of start-up business:
    B=100, E=20，ROE=20%，PE=120/20=6；
  - two years later, assume a PE fund chip in based on the PE=10 and E=24 for 50% share

$$\frac{P_3}{P_1} = \frac{P_3/E_3}{P_1/E_1} \times \frac{ROE_3}{ROE_1} \times \frac{B_2}{B_0} = \frac{10}{6} \times \frac{20\%}{20\%} \times \frac{192}{100} = 3.2$$

» The value at the end of second year is (100+20+24+240)*0.2*10=768，
in which 384 belongs to the original shareholders

» Conclusion: value created by the combination of valuation, earnings and financing

The Value of Growth

- The growth of value comes from
  - – higher earnings
    - » enhance efficiency
    - » leverage
  - –higher valuation
  - –favorable equity financing

- Companies can increase their asset through capital market (equity, debt, loan)

Growth Path

- growth path of a comany
  - value created by valuation, earings and leverage

$$\frac{P_2}{P_1} = \frac{P_2 / E_2}{P_1 / E_1} \times \frac{ROE_2}{ROE_1} \times \frac{B_1}{B_0}$$

  - DuPont decomposition：

$$ROE_1 = \frac{E_1}{B_0} = \frac{E_1}{S_1} \times \frac{S_1}{A_0} \times \frac{A_0}{B_0} \qquad ROE = ROA \times \frac{A_0}{B_0}$$

$\frac{E_1}{S_1}$: sales margin； $\frac{S_1}{A_0}$: asset turnover rate； $\frac{A_0}{B_0}$: equity multiplier

净资产收益率 = 销售净利率*总资产周转率*权益乘数

= （净利润/营业收入）*（营业收入/总资产）*（总资产/净资产）

ROE = sales margin * asset turnover rate * equity multiplier

Maotai: sales margin 50% (gross margin 90%); asset turnover rate 0.5-0.6; equity multiplier 1.32

Walmart: sales margin 3%; asset turnover rate 2.4; equity multiplier 2.69

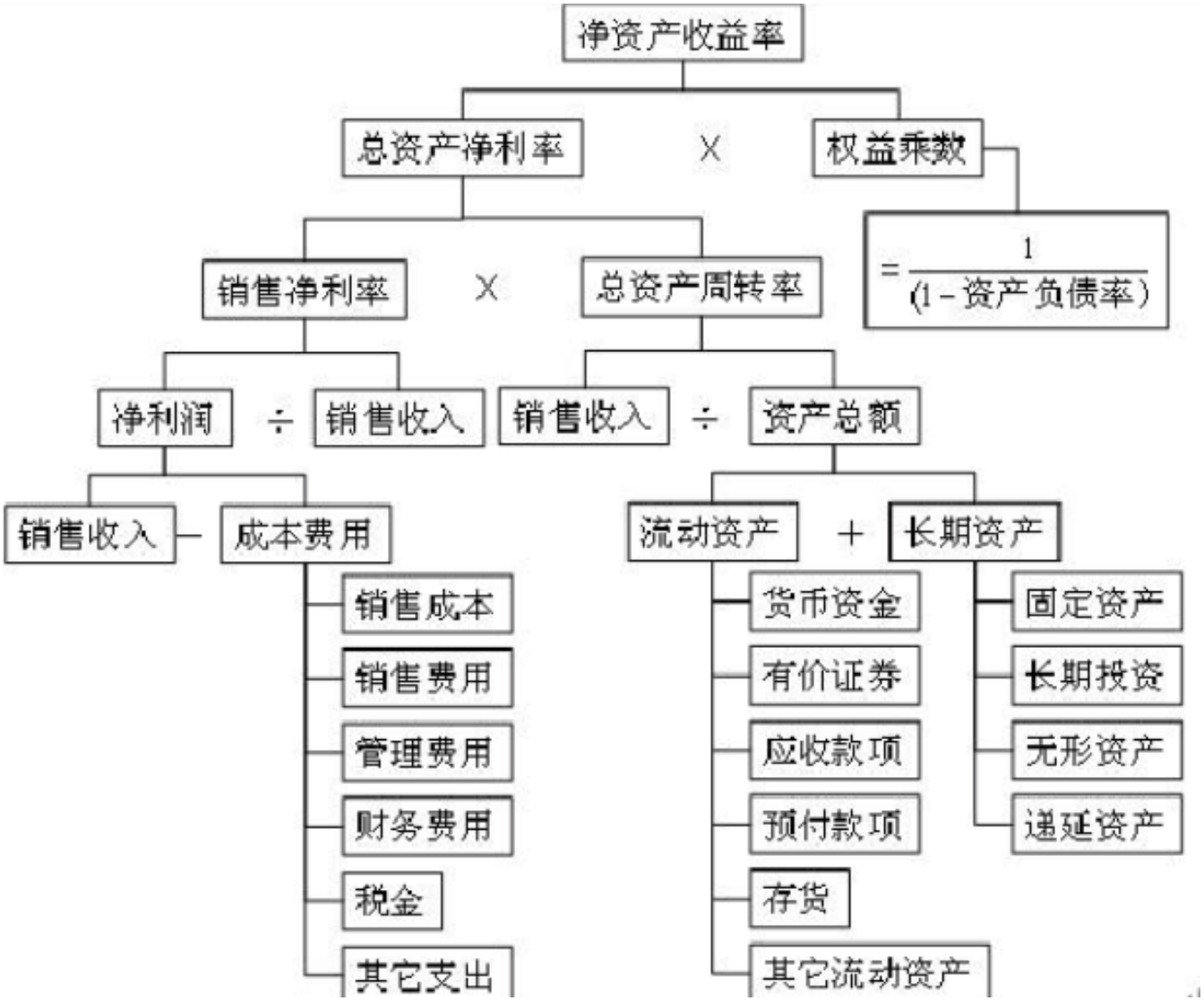
Bank of China: sales margin 33%; asset turnover rate 0.03; equity multiplier 12-13

capital turnover rate

sales margin Equity multiplier

分类	高ROE模式	企业	ROE	销售净利润率	总资产周转率	权益乘数
模式一	高净利率	中海地产	20.86%	28%	28%	2.67
模式二	高周转	万科地产	19.61%	13.50%	32%	4.54
模式三	高杠杆	泰禾集团	20.15%	11.50%	25%	7.01

DuPont Decomposition



Case study: Zhong Lian Zhong Gong

The firm is the world's fifth largest Engineering Machinery with total asset of 131 billion RMB.
The stock price decline from 15 yuan/share in 2011 to 9 yuan/share in 2013



Case study: Zhong Lian Zhong Gong

中联重科2011年杜邦分析

Return on Equity
25.02%

Return on Assets × **Equity Multiplier**
12.45% × 2.01

Net Profit Margin × **Asset Turnover Rate**
18.08% × 0.69

Net Income ÷ **Revenue**
8,375,299,690.70 ÷ 46,322,580,713.20

Revenue	-	Expenses	+	Other Revenue or Expenses	-	Income Tax Expense
46,322,580,713.20		36,606,253,079.63		88,104,138.75		1,429,132,081.62

Cost of Revenue	+	Selling Expenses	+	Administrative Expenses	+	Financing Expenses	+	Business Taxes and Surcharges
31,316,026,319.85		3,159,739,323.20		1,755,880,397.95		110,979,689.01		263,627,349.62

Revenue ÷ **Average Balance of Assets**
46,322,580,713.20 ÷ 67,288,523,588.64

Current assets + **Non-current assets**

Current assets		Non-current assets	
Beginning Balance:	44,366,461,923.86	Beginning Balance:	18,628,813,574.23
Ending Balance:	49,015,547,831.41	Ending Balance:	22,566,223,847.78

Cash and Cash Equivalents		Long-term Equity Investment	
Beginning Balance:	20,519,895,456.36	Beginning Balance:	94,487,195.88
Ending Balance:	17,744,514,431.30	Ending Balance:	115,055,154.45

Advance Payment		Fixed Asset	
Beginning Balance:	669,775,009.96	Beginning Balance:	3,657,857,080.31
Ending Balance:	733,248,385.84	Ending Balance:	4,101,571,488.87

Accounts Receivable		Intangible Assets	
Beginning Balance:	6,947,232,058.99	Beginning Balance:	2,375,049,313.18
Ending Balance:	11,658,065,230.17	Ending Balance:	2,605,917,963.98

Inventory		Others	
Beginning Balance:	8,678,146,510.00	Beginning Balance:	12,501,419,984.86
Ending Balance:	9,655,357,242.28	Ending Balance:	15,743,679,240.48

Others	
Beginning Balance:	7,551,412,888.55
Ending Balance:	9,224,362,541.82

中联重科2013年杜邦分析

Return on Equity
9.43%

×

Return on Assets
4.43%

×

Equity Multiplier
2.13

Net Profit Margin
10.25%

×

Asset Turnover Rate
0.43

Net Income	÷	Revenue
3,951,881,334.27		38,541,775,254.36

Revenue	÷	Average Balance of Assets
38,541,775,254.36		89,255,808,859.89

Revenue	-	Expenses	+	Other Revenue or Expenses	-	Income Tax Expense
38,541,775,254.36		33,019,113,032.91		-12,325,750.79		570,029,454.77

Cost of Revenue	+	Selling Expenses	+	Administrative Expenses	+	Financing Expenses	+	Business Taxes and Surcharges
27,300,089,843.06		3,631,169,614.06		1,953,446,392.73		-136,448,074.66		270,855,257.72

Current assets		+	Non-current assets	
Beginning Balance:	67,029,570,000.00		Beginning Balance:	21,944,890,000.00
Ending Balance:	69,529,958,513.30		Ending Balance:	20,007,199,206.47

Cash and Cash Equivalents		Long-term Equity Investment	
Beginning Balance:	23,206,160,000.00	Beginning Balance:	330,060,000.00
Ending Balance:	20,072,218,953.24	Ending Balance:	337,566,831.96

Advance Payment		Fixed Asset	
Beginning Balance:	838,310,000.00	Beginning Balance:	4,569,360,000.00
Ending Balance:	524,633,111.56	Ending Balance:	5,376,849,583.90

Accounts Receivable		Intangible Assets	
Beginning Balance:	18,900,350,000.00	Beginning Balance:	2,731,120,000.00
Ending Balance:	27,806,307,708.90	Ending Balance:	2,930,977,945.18

Inventory		Others	
Beginning Balance:	11,732,530,000.00	Beginning Balance:	14,314,350,000.00
Ending Balance:	8,747,198,355.13	Ending Balance:	11,361,804,845.43

Others	
Beginning Balance:	12,352,220,000.00
Ending Balance:	12,379,600,384.47

Analysis

1. ROE decline from 25.02% to 9.43%, due to the decline of ROA
2. ROA=销售净利率(sales margin) *总资产周转率(capital turnover rate)
   In 2011, ROA=18.08%*0.69; In 2013, ROA=10.25%*0.43
   both sales margin and capital turnover rate decline
3. 销售净利率(sales margin)=净利润/营业收入 (sales margin=net profit/sales)
   2011销售净利率(sales margin)= 8,173,341,830 / 46,322,580,713 =18.08%
   2013销售净利率(sales margin)= 3,951,881,334 / 38,541,775,254 =10.25%
   Both sales and net profit decline
4. 成本费用(COGS and SG&A)
   In 2011, 成本费用= 36,606,253,079; in 2013, 成本费用= 33,019,113,032
   sales collapse while cost did not go down much. Competition!
5. The capital turnover rate goes down from 0.69 in 2011 to 0.43 in 2013
6. Capital turnover rate=sales/total asset
   Sales decline, while总资产平均余额 increase substantially
7. Why? (Account receivable)

2011

Average Balance of Assets

67,288,523,588.64

Current assets	
Beginning Balance:	44,366,461,923.86
Ending Balance:	49,015,547,831.41

Cash and Cash Equivalents	
Beginning Balance:	20,519,895,456.36
Ending Balance:	17,744,514,431.30

Advance Payment	
Beginning Balance:	669,775,009.96
Ending Balance:	733,248,385.84

Accounts Receivable	
Beginning Balance:	6,947,232,058.99
Ending Balance:	11,658,065,230.17

Inventory	
Beginning Balance:	8,678,146,510.00
Ending Balance:	9,655,357,242.28

Others	
Beginning Balance:	7,551,412,888.55
Ending Balance:	9,224,362,541.82

+

Non-current assets	
Beginning Balance:	18,628,813,574.23
Ending Balance:	22,566,223,847.78

Long-term Equity Investment	
Beginning Balance:	94,487,195.88
Ending Balance:	115,055,154.45

Fixed Asset	
Beginning Balance:	3,657,857,080.31
Ending Balance:	4,101,571,488.87

Intangible Assets	
Beginning Balance:	2,375,049,313.18
Ending Balance:	2,605,917,963.98

Others	
Beginning Balance:	12,501,419,984.86
Ending Balance:	15,743,679,240.48

2013

Average Balance of Assets

89,255,808,859.89

Current assets	
Beginning Balance:	67,029,570,000.00
Ending Balance:	69,529,958,513.30

Cash and Cash Equivalents	
Beginning Balance:	23,206,160,000.00
Ending Balance:	20,072,218,953.24

Advance Payment	
Beginning Balance:	838,310,000.00
Ending Balance:	524,633,111.56

Accounts Receivable	
Beginning Balance:	18,900,350,000.00
Ending Balance:	27,806,307,708.90

Inventory	
Beginning Balance:	11,732,530,000.00
Ending Balance:	8,747,198,355.13

Others	
Beginning Balance:	12,352,220,000.00
Ending Balance:	12,379,600,384.47

+

Non-current assets	
Beginning Balance:	21,944,890,000.00
Ending Balance:	20,007,199,206.47

Long-term Equity Investment	
Beginning Balance:	330,060,000.00
Ending Balance:	337,566,831.96

Fixed Asset	
Beginning Balance:	4,569,360,000.00
Ending Balance:	5,376,849,583.90

Intangible Assets	
Beginning Balance:	2,731,120,000.00
Ending Balance:	2,930,977,945.18

Others	
Beginning Balance:	14,314,350,000.00
Ending Balance:	11,361,804,845.43

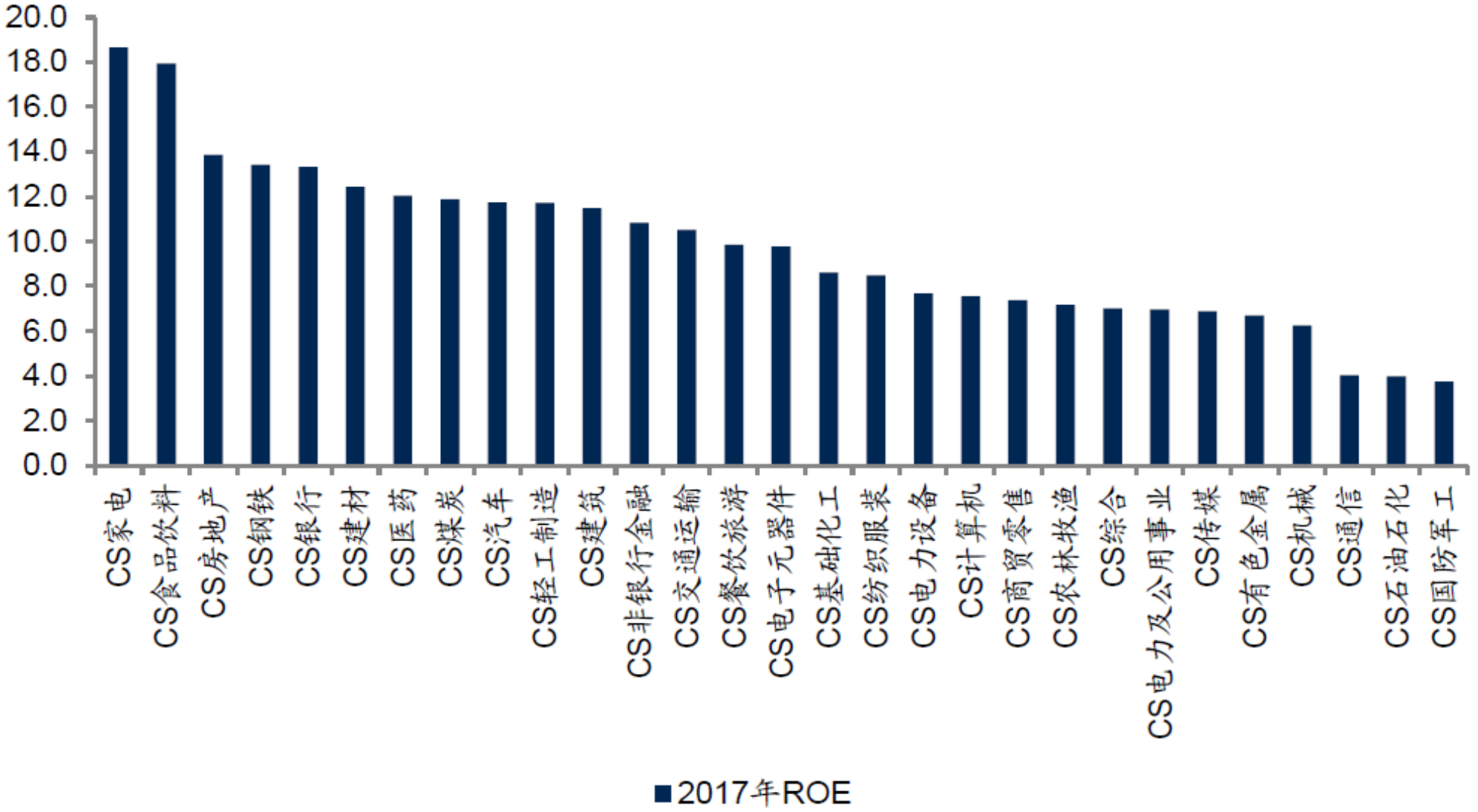
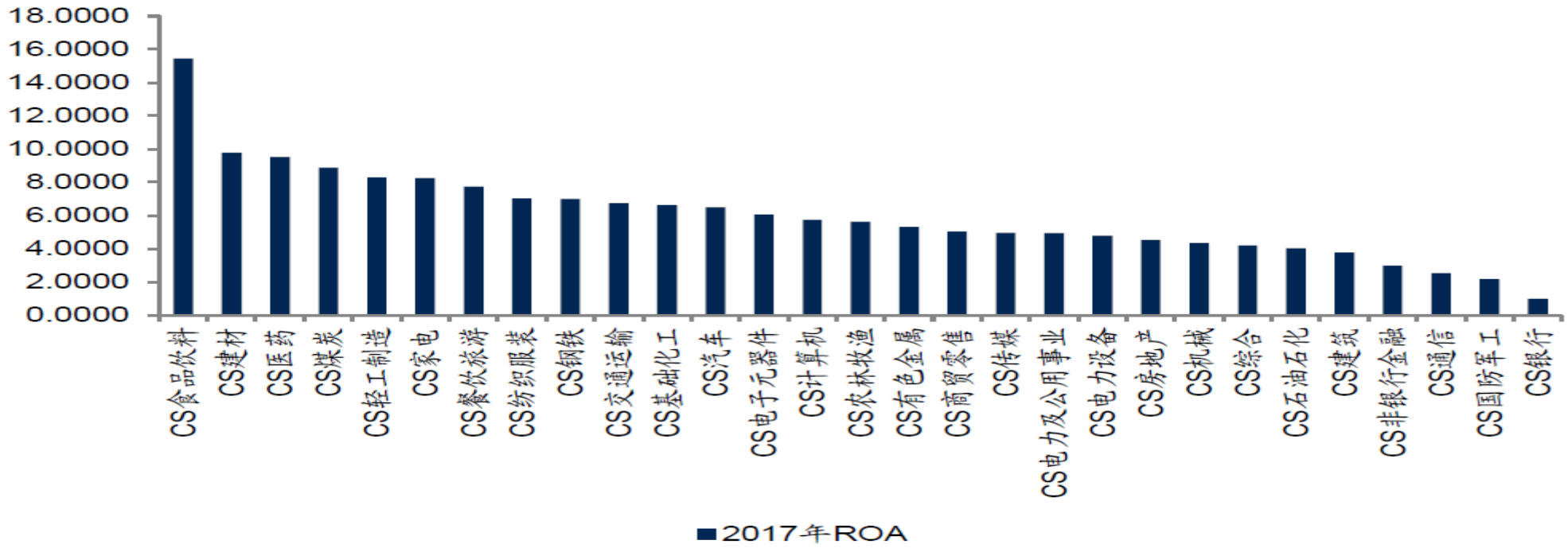
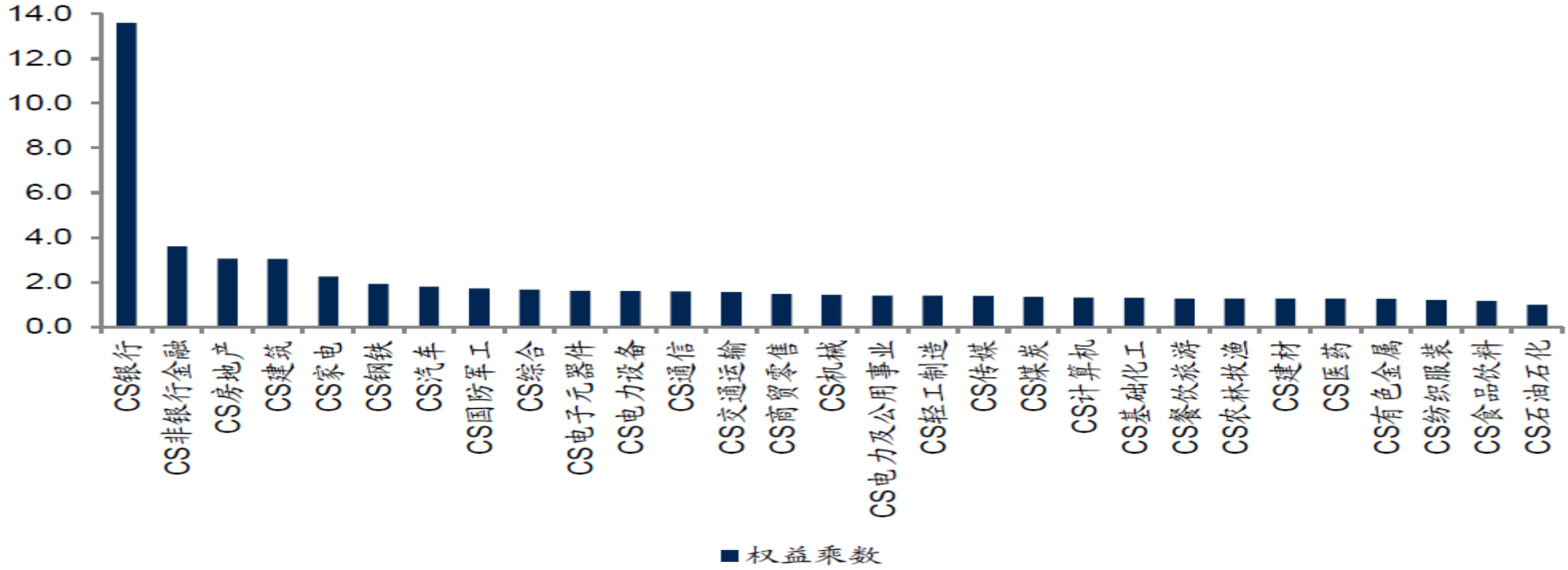

20.0
18.0
16.0
14.0
12.0
10.0
8.0
6.0
4.0
2.0
0.0
CS家电
CS食品饮料
CS房地产
CS钢铁
CS银行
CS建材
CS医药
CS煤炭
CS汽车
CS轻工制造
CS建筑
CS非银行金融
CS交通运输
CS餐饮旅游
CS电子元器件
CS基础化工
CS纺织服装
CS电力设备
CS计算机
CS商贸零售
CS农林牧渔
CS综合
CS电力及公用事业
CS传媒
CS有色金属
CS机械
CS通信
CS石油石化
CS国防军工
2017年ROE

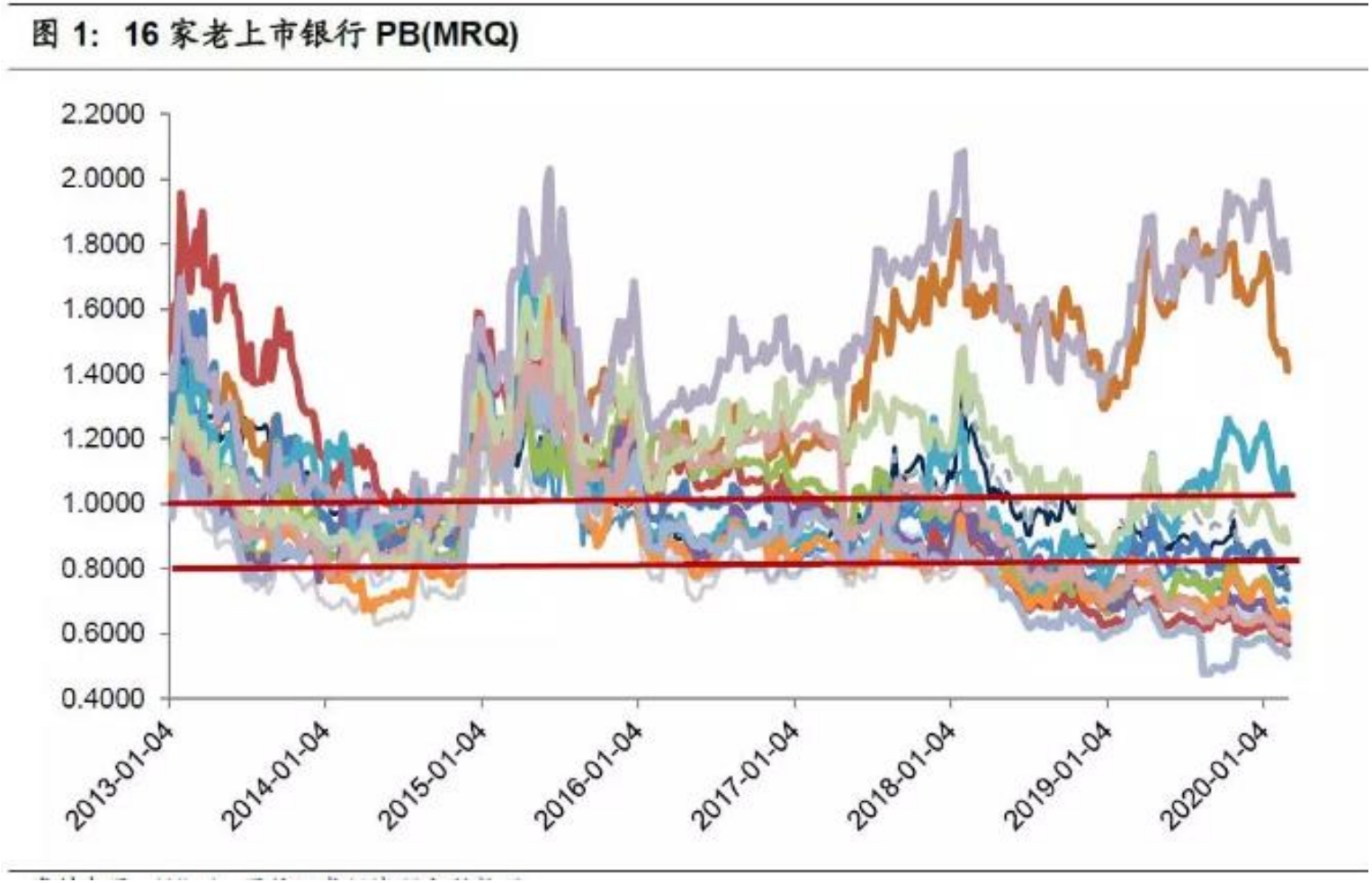
图 4：银行板块 ROA 非常低





Another example of using ROE/ROA

【国信银行】“让利”导致银行股低估值吗？



Bank valuation is at historical low

The starting point of the analysis:

$$PB = \frac{ROE \cdot d}{(r - ROE) + ROE \cdot d}$$

Where does it come from?

$$\frac{P}{B} = \frac{ROE\ (1 - b)}{r - b \times ROE}$$ Remember?

We often compare PB with 1. According to the above formula, when r>ROE, PB<1.

$$PB = \frac{ROE \cdot d}{(r - ROE) + ROE \cdot d} \Rightarrow ROE = \frac{PB \cdot r}{d - PB \cdot d + PB}$$

Given PB=0.8, dividend payout ratio d=25%, if we assume r=10%=〉 ROE=9.4%

Compare it to the current banking industry average ROE=12%, ROE should go down by 2.6%, ROA should go from 1% to 78bps(given the leverage is 12)

22bps decline of ROA is significant, whether it is reasonable or not depends on one's insight of the banking industry. According to the report this translate into the loan interest rate cut by 44bps, which is unlikely. Another perspective is that in other countries, ROA is around 1%

表 1：估值表

收盘日：	港元汇率	收盘价	普通股 ROE			归母净利润同比			PB			PE		
2020/2/28	0.8987	（元）	2018A	2019E	2020E	2018A	2019E	2020E	2018A	2019E	2020E	2018A	2019E	2020E
601398.SH	工商银行	5.30	13.7%	13.0%	12.5%	4.1%	5.0%	5.1%	0.84	0.77	0.70	6.4	6.1	5.8
601288.SH	农业银行	3.41	13.4%	12.5%	12.2%	5.1%	5.8%	5.5%	0.75	0.69	0.63	6.0	5.7	5.4
601988.SH	中国银行	3.55	12.0%	11.3%	11.1%	4.5%	3.9%	4.9%	0.69	0.64	0.59	6.0	5.8	5.5
601328.SH	交通银行	5.20	11.5%	11.2%	10.6%	4.9%	4.5%	2.8%	0.60	0.56	0.52	5.4	5.2	5.0
1658.HK	邮储银行	4.42	13.2%	12.7%	12.6%	9.7%	15.1%	13.2%	0.90	0.77	0.70	7.1	6.6	5.8
—														
600036.SH	招商银行	34.20	16.6%	16.6%	16.9%	14.8%	14.3%	14.4%	1.70	1.51	1.33	10.9	9.5	8.3
601166.SH	兴业银行	17.29	14.2%	13.8%	13.4%	6.0%	8.6%	8.2%	0.82	0.73	0.66	6.1	5.6	5.1
600016.SH	民生银行	5.72	13.0%	12.2%	11.7%	1.0%	3.9%	4.8%	0.61	0.56	0.51	4.8	4.6	4.4
600000.SH	浦发银行	10.85	13.0%	12.8%	12.4%	3.1%	10.7%	6.6%	0.72	0.65	0.59	5.9	5.3	5.0
601998.SH	中信银行	5.43	11.2%	10.5%	10.1%	4.6%	4.2%	3.1%	0.66	0.61	0.57	6.2	5.9	5.7
000001.SZ	平安银行	14.50	11.4%	11.2%	10.9%	7.0%	13.6%	13.9%	1.28	1.03	0.94	11.8	10.3	9.0
601818.SH	光大银行	3.78	11.7%	11.4%	11.1%	6.7%	6.0%	5.6%	0.68	0.63	0.58	6.2	5.8	5.5
600015.SH	华夏银行	6.95	12.3%	10.1%	9.1%	5.2%	0.9%	1.2%	0.54	0.50	0.46	5.3	5.3	5.2
—														
601169.SH	北京银行	5.11	11.6%	11.2%	11.1%	6.8%	6.2%	6.9%	0.62	0.57	0.53	5.6	5.3	4.9
601009.SH	南京银行	7.76	16.9%	16.6%	16.8%	14.5%	13.2%	13.0%	0.97	0.86	0.76	6.2	5.4	4.8
002142.SZ	宁波银行	25.16	19.0%	17.3%	16.9%	19.9%	18.5%	16.4%	2.14	1.67	1.45	12.9	10.9	9.3
600919.SH	江苏银行	6.38	12.4%	13.1%	13.5%	10.0%	15.5%	13.6%	0.72	0.65	0.59	6.1	5.2	4.6
601077.SH	渝农商行	5.74	13.5%	13.9%	13.3%	1.4%	19.1%	16.8%	0.93	0.73	0.66	7.2	6.0	5.2
—														
603323.SH	苏农银行	4.62	9.2%	8.6%	8.8%	9.7%	15.9%	13.7%	0.91	0.72	0.67	10.4	9.0	7.9
002839.SZ	张家港行	5.39	9.2%	9.5%	10.1%	9.4%	14.0%	14.5%	1.03	0.96	0.89	11.7	10.2	8.9
601128.SH	常熟银行	7.94	13.3%	11.8%	12.3%	17.5%	16.6%	19.4%	1.70	1.32	1.21	14.6	12.3	10.3
	平均		***12.9%***	***12.4%***	***12.2%***	***8.0%***	***10.3%***	***9.7%***	***0.94***	***0.82***	***0.74***	***7.8***	***7.0***	***6.3***

06 投资建议

综上，我们认为，至少对部分盈利能力仍然较高的优质个股而言，其ROA降至类似0.78%这样的低位，是非常小概率的。而它们目前交易在0.8倍以下的估值水平，存在明显低估，投资价值显著。股价的触发器可能在于后续半年甚至一年内，实际情况表明疫情或所谓的让利政策未对其经营产生重大影响，比如贷款利率、其他业务收入均较平稳。因此，目前市场在近期非理性波动下形成的极低估值水平，可能是一个良好介入时机。

因此，我们维持行业"超配"评级不变，并重点推荐估值极低、盈利稳定的大型国有行，包括工商银行、农业银行、邮储银行等。同时也推荐其他基本面优异的中小银行，包括宁波银行、常熟银行等。

07 风险提示

若宏观经济大幅下行，可能从多方面影响银行业，比如经济下行时期货币政策宽松对净息差可能产生负面影响、宏观经济下行时期借款人偿债能力下降可能会对银行资产质量产生不利影响等。

DCF model：calculating equity value using FCFE

- Equity present value
  - discount rate
    - » same as calculating dividend
  - cash flow
    - » free cash flow of equity（FCFE）=

      net income + depreciation - capital expenditure -

      change of working capital (account receivable, inventory, etc.)
  - final value
    - » assuming the company has a permanent steady growth rate

      the final value at n= $FCFE/(r-g)$

Large capital expenditure and significant change of working capital could be bad signal (account receivable and inventory increase indicate low quality profit)

Pricing Model of Company

- Company with stable growth

$$V = \frac{FCFF_1}{WACC - g_n}$$

$$\text{WACC} = r_d \times \frac{D}{D+E} + r_e \times \frac{E}{D+E}$$

2-stage Pricing Model

$$V_0 = \sum_{1}^{n} \frac{FCFF_t}{(1+WACC_{hg})} + \frac{V_n}{(1+WACC_{hg})^n}$$

$$V_0 = \frac{FCFF_0 \times (1+g) \times \left[1 - \frac{(1+g)^n}{(1+WACC_{hg})^n}\right]}{(k_{hg} - g)} + \frac{FCFF_0 \times (1+g)^n \times (1+g_n)}{(k_{st} - g_n) \times (1+k_{hg})^n}$$

g = the growth rate of previous n years

k_{hg} = the discounted rate of high growth period

k_{st} = the discounted rate of stable growth

g_n = the stable growth rate after n years